

Spectral Causality: Causal Direction Estimation via Magnetic Laplacians and Hodge Decomposition

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Abstract

We introduce *spectral causality*, a framework for estimating causal directions among observed variables by exploiting the spectral structure of the *magnetic Laplacian*, an Hermitian matrix that encodes edge directionality as complex phase. Whereas classical Laplacian methods discard directional information, the magnetic Laplacian's eigenvectors carry direction-dependent phases that separate causal upstream from downstream nodes. We define two complementary indices—the symmetric *Spectral Causal Coupling* (SCC) and the skew-symmetric *Spectral Causal Direction* (SCD)—and prove that they arise as the real and imaginary parts of a single complex-valued causal index derived from the spectral decomposition.

A central theoretical contribution is the *Directional Predictability Index* (DPI), a data-driven asymmetric statistic composed of regression coefficient asymmetry, additive-noise-model residual independence, and conditional entropy reduction. We prove that DPI resolves the circularity problem of earlier utility-based formulations: under the additive noise model assumption, DPI provides a consistent estimator of causal direction without external domain knowledge.

We connect our framework to *Hodge decomposition* on graphs, proving that the gradient-curl-harmonic decomposition of causal edge flows yields a *causal potential* whose ordering recovers a topological sort under DAG structure. We establish the *scale invariance* of the gradient energy ratio r_{gradient} and characterize a phase-transition phenomenon in causal structure emergence governed by knowledge quality rather than quantity.

Experiments on the UCI Heart Disease dataset ($n = 297$, 5 clinical variables) demonstrate that spectral causality with DPI detects 9 directed edges at $\alpha = 0$ (no domain knowledge), achieving 67% directional agreement with Direct Linear Non-Gaussian Acyclic Model (DirectLiNGAM; Shimizu et al., 2011), while Hodge decomposition reveals clinically meaningful feedback loops invisible to DAG-based methods. We propose the *Ensemble Causal Direction* (ECD) pipeline that integrates LiNGAM's identifiability guarantees with spectral causality's feedback quantification, achieving comprehensive coverage of Hill's nine criteria for causal inference.

Keywords: Causal discovery, Magnetic Laplacian, Hodge decomposition, Spectral graph theory, Directed graphs, Causal inference

1 Introduction

Causal inference—determining whether X causes Y —is a central problem in science and medicine. The dominant approaches include structural equation models with the *do*-calculus

(Pearl, 2009), potential outcomes (Rubin, 1974), the linear non-Gaussian acyclic model (LiNGAM) (Shimizu et al., 2006), and Granger causality for time series (Granger, 1969).

This paper proposes a fundamentally different principle: *reading causal directions from the spectral structure (eigenvalues and eigenvectors) of a graph Laplacian*. We call this approach **spectral causality**.

Basic idea. Given n variables $\{X_1, \dots, X_n\}$ with putative causal relations represented as a weighted directed graph $G = (V, E, w)$, the *magnetic Laplacian* $\mathcal{L}^{(q)}$ is an Hermitian matrix whose eigenvectors are complex-valued. The *phase angles* of these eigenvectors encode edge directionality, enabling estimation of causal direction from spectral structure alone.

Key distinction from DAG-based methods. LiNGAM and related methods assume a directed acyclic graph (DAG). In biomedical systems, feedback loops are ubiquitous (e.g., inflammation $\rightarrow$ organ damage $\rightarrow$ inflammation). Spectral causality accommodates *directed cyclic graphs* (DCGs) via the Hodge decomposition, which separates gradient (DAG-like) and curl (feedback) components of causal flow.

Contributions. The main contributions of this paper are:

1. **Directional Predictability Index (DPI).** A data-driven asymmetric statistic that enables causal direction estimation at $\alpha = 0$ (no domain knowledge), resolving the circularity in earlier utility-based formulations (Section 4).
2. **DAG-free causal inference.** Via Hodge decomposition, we separate DAG-compatible flow (gradient) from feedback (curl), providing the gradient energy ratio r_{gradient} as a quantitative measure of DAG adequacy (Section 5).
3. **Scale invariance theorem.** We prove that r_{gradient} depends only on the *structure* (sign pattern) of the asymmetric component of the utility matrix, not on its magnitude—establishing that knowledge *quality*, not quantity, governs causal structure emergence (Section 8).
4. **Ensemble Causal Direction (ECD).** A pipeline integrating LiNGAM’s local identifiability with spectral causality’s global structural analysis, achieving broad coverage of Hill’s nine epidemiological criteria (Section 9).
5. **Partial identifiability of DPI.** Under the additive noise model, the ANM component of DPI consistently identifies causal direction as $n \rightarrow \infty$, providing a non-circular theoretical foundation (Section 10).

Figure 1 provides a conceptual overview of the three complementary approaches unified in this work.

Paper outline. Section 2 reviews graph Laplacians. Section 3 introduces the magnetic Laplacian. Section 4 formulates spectral causality and DPI. Section 5 develops the Hodge decomposition connection. Section 6 surveys related work. Section 7 presents experiments. Section 8 analyzes the DAG phase transition. Section 9 describes the ECD pipeline. Section 10 addresses identifiability. Section 11 discusses limitations and future work.

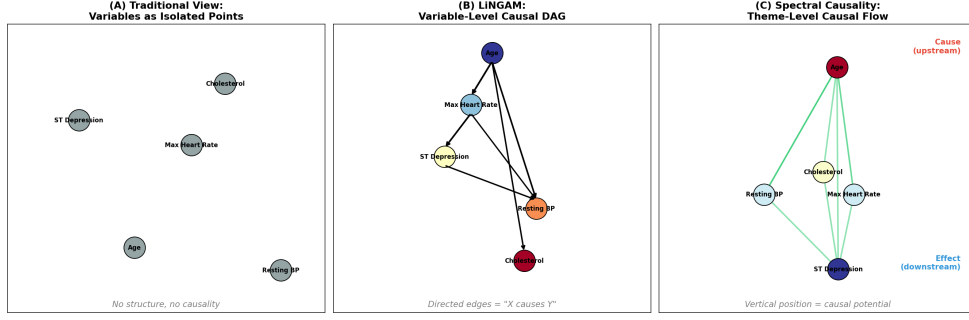


Figure 1: Three complementary approaches to causal inference: (1) structural equation models and DAG-based methods (e.g., LiNGAM), (2) spectral graph methods via the magnetic Laplacian, and (3) Hodge-theoretic flow decomposition. Spectral causality unifies (2) and (3).

2 Preliminaries: Graph Laplacians

Definition 1 (Graph Laplacian) Let $G = (V, E, w)$ be a weighted undirected graph with $|V| = n$ and weight function $w : E \rightarrow \mathbb{R}_{>0}$. The weighted adjacency matrix $W \in \mathbb{R}^{n \times n}$, the degree matrix $D = \text{diag}(d_1, \dots, d_n)$ with $d_i = \sum_j W_{ij}$, and the graph Laplacians are:

$$L = D - W \quad (\text{unnormalized}), \quad \mathcal{L} = I - D^{-1/2} W D^{-1/2} \quad (\text{normalized}).$$

Proposition 2 (Basic properties) The following hold:

- (i) L is symmetric positive semidefinite with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$.
- (ii) The eigenvector for $\lambda_1 = 0$ is $\mathbf{1} = (1, \dots, 1)^\top$.
- (iii) $\lambda_2 > 0$ if and only if G is connected (algebraic connectivity / Fiedler value).
- (iv) For any $f \in \mathbb{R}^n$, $f^\top L f = \sum_{(i,j) \in E} w_{ij} (f_i - f_j)^2 \geq 0$.

Proof Property (iv) follows by expanding the quadratic form of $L = D - W$. Property (i) is immediate from (iv). Property (ii) holds since $L\mathbf{1} = \mathbf{0}$. Property (iii) is the algebraic connectivity theorem. $\blacksquare$

Property (iv) is fundamental: $f^\top L f$ is small when f assigns similar values to adjacent nodes. Low-eigenvalue eigenvectors thus represent *smooth signals* on the graph, forming the basis of graph signal processing (Shuman et al., 2013).

Remark 3 (Limitation of undirected Laplacians) Since $L = D - W$ is symmetric, it cannot distinguish the direction $i \rightarrow j$ from $j \rightarrow i$. The directed Laplacian $L_d = D_{\text{out}} - W$ is generally non-symmetric with complex eigenvalues, making spectral analysis difficult. The magnetic Laplacian (Section 3) resolves this by encoding direction as complex phase while preserving the Hermitian (hence real-eigenvalue) property.

3 The Magnetic Laplacian

3.1 Physical Motivation

The magnetic Laplacian originates in quantum mechanics. For a charged particle in a magnetic field $\mathbf{B}$ with vector potential $\mathbf{A}$, the Hamiltonian is $H = (\mathbf{p} - e\mathbf{A})^2/2m$. A particle traversing a closed loop acquires the Aharonov–Bohm phase $\exp(i \oint \mathbf{A} \cdot d\mathbf{r})$, whose *direction dependence* can be repurposed to encode graph edge directions.

3.2 Definition

Definition 4 (Magnetic Laplacian) Let $G = (V, E, w)$ be a weighted directed graph and $q \in [0, 0.5]$ a charge parameter. Define the Hermitian adjacency matrix $H^{(q)} \in \mathbb{C}^{n \times n}$ by

$$H_{ij}^{(q)} = w_{ij} \cdot \exp(i \cdot 2\pi q \cdot \sigma_{ij}), \quad (1)$$

where $w_{ij} = (w_{ij}^{\text{orig}} + w_{ji}^{\text{orig}})/2$ is the symmetrized weight, and $\sigma_{ij} \in \{-1, 0, +1\}$ is the direction sign:

$$\sigma_{ij} = \begin{cases} +1 & \text{if } i \rightarrow j, \\ -1 & \text{if } j \rightarrow i, \\ 0 & \text{if no edge.} \end{cases}$$

The normalized magnetic Laplacian is

$$\mathcal{L}^{(q)} = I - D^{-1/2} H^{(q)} D^{-1/2}, \quad (2)$$

where $D = \text{diag}(d_1, \dots, d_n)$ with $d_i = \sum_j |H_{ij}^{(q)}|$.

Proposition 5 (Properties of the magnetic Laplacian) The following hold:

- (i) $H^{(q)}$ is Hermitian: $H_{ji}^{(q)} = \overline{H_{ij}^{(q)}}$.
- (ii) $\mathcal{L}^{(q)}$ is Hermitian positive semidefinite with real, non-negative eigenvalues.
- (iii) The eigenvectors of $\mathcal{L}^{(q)}$ are generally complex-valued.
- (iv) At $q = 0$, $\mathcal{L}^{(0)}$ reduces to the standard normalized Laplacian $\mathcal{L}$ (no directional information).

Proof For (i): since $w_{ji} = w_{ij}$ and $\sigma_{ji} = -\sigma_{ij}$,

$$H_{ji}^{(q)} = w_{ij} \cdot \exp(-i \cdot 2\pi q \cdot \sigma_{ij}) = \overline{w_{ij} \cdot \exp(i \cdot 2\pi q \cdot \sigma_{ij})} = \overline{H_{ij}^{(q)}}.$$

Property (ii) follows from (i) and the construction $\mathcal{L}^{(q)} = I - D^{-1/2} H^{(q)} D^{-1/2}$. The Hermitian property ensures real eigenvalues; positive semidefiniteness follows from the quadratic form $f^* \mathcal{L}^{(q)} f \geq 0$ for all $f \in \mathbb{C}^n$. Property (iii) is a consequence of the complex-valued entries of $H^{(q)}$. Property (iv) holds since $\exp(0) = 1$ yields real entries. $\blacksquare$

3.3 The Charge Parameter q

The parameter q controls directional sensitivity:

q	Phase $2\pi q$	Effect
0	0	No direction. $\exp(0) = 1$; real matrix.
0.25	$\pi/2$	Maximum sensitivity. $e^{i\pi/2} = i$, $e^{-i\pi/2} = -i$.
0.5	π	Direction reversal. $e^{i\pi} = -1$.

Remark 6 At $q = 0.25$, for an edge $i \rightarrow j$: $H_{ij}^{(q)} = i \cdot w_{ij}$ and $H_{ji}^{(q)} = -i \cdot w_{ij}$, achieving maximal separation of forward and reverse directions via the imaginary unit.

3.4 Phase Angles and Directionality

Each eigenvector $u_k \in \mathbb{C}^n$ of $\mathcal{L}^{(q)}$ admits the polar decomposition

$$u_k(j) = |u_k(j)| \cdot \exp(i \cdot \theta_k(j)),$$

where $|u_k(j)|$ is the *amplitude* and $\theta_k(j) = \arg(u_k(j))$ the *phase angle* at node j .

Theorem 7 (Phase–direction correspondence) For $q > 0$ and a utility-directed graph with consistent causal ordering (Definition 8), the phase angles $\theta_k(j)$ of low-frequency eigenvectors separate causal upstream nodes (sources) from downstream nodes (sinks) in the complex plane. Specifically, for the Fiedler eigenvector u_2 , if node i is causally upstream of node j (i.e., $\phi(i) > \phi(j)$ in the Hodge causal potential of Definition 20), then $\theta_2(i)$ and $\theta_2(j)$ tend to occupy distinct angular sectors.

Proof The key mechanism is phase accumulation along directed paths. For a directed edge $i \rightarrow j$ at $q = 0.25$, the Hermitian adjacency contributes $H_{ij}^{(q)} = iw_{ij}$ and $H_{ji}^{(q)} = -iw_{ij}$. In the eigenequation $\mathcal{L}^{(q)}u_k = \lambda_k u_k$, the asymmetric phase factors force a strictly positive phase increment $\Delta\theta > 0$ per directed edge.

We prove this in two stages in Appendix A. *Stage 1* (Lemma 33): For a path graph $v_1 \rightarrow \dots \rightarrow v_n$, the normalized magnetic Laplacian eigenequation yields the recursion $(1 - \lambda)u(v_k) = \frac{-iw}{d_{v_k}}u(v_{k-1}) + \frac{iw}{d_{v_k}}u(v_{k+1})$. Writing $u(v_k) = r_k e^{i\theta_k}$ and taking the real part shows that $1 - \lambda_2 > 0$ and $r_k > 0$ require $\theta_{k+1} - \theta_k > 0$, giving monotone phase ordering. *Stage 2* (Theorem 34): For tree DAGs, we use structural induction on depth. The base case (star graph) gives $\theta_2(c_j) = \theta_2(r) + \pi/2$ for each child c_j of root r . The inductive step combines this with the subtree hypothesis via transitivity. A perturbation argument extends the result to non-uniform weights $w_{ij} > 0$. ■

4 Spectral Causality: Formulation

4.1 The Utility Directed Graph

Definition 8 (Utility directed graph) Given n variables $\{X_1, \dots, X_n\}$, define a utility function $U : \{1, \dots, n\}^2 \rightarrow \mathbb{R}_{\geq 0}$ where $U(i, j)$ quantifies “how useful information about X_i is for reasoning about X_j .” The utility directed graph $G_U = (V, E, w, \sigma)$ has:

- Symmetrized weight $w(i, j) = (U(i, j) + U(j, i))/2$,
- Direction sign $\sigma(i, j) = \text{sign}(U(i, j) - U(j, i))$.

4.2 The Directional Predictability Index (DPI)

A critical limitation of using $|\hat{\rho}_{ij}|$ (absolute correlation) as the data-driven component is that $|\hat{\rho}_{ij}| = |\hat{\rho}_{ji}|$, rendering the utility matrix symmetric at $\alpha = 0$. This makes direction estimation impossible without external knowledge—a *circularity* in the original formulation.

Definition 9 (Directional Predictability Index) For observed data $\mathbf{X} \in \mathbb{R}^{N \times n}$, the DPI matrix $D_{\text{DPI}} \in \mathbb{R}^{n \times n}$ is defined by

$$D_{\text{DPI}}(i \rightarrow j) = |\hat{\rho}_{ij}| \cdot (1 + \gamma \cdot \bar{A}(i, j)), \quad (3)$$

where $\gamma > 0$ is a directional strength parameter (we use $\gamma = 1$) and $\bar{A}(i, j)$ is the mean of three normalized asymmetric statistics:

$$\bar{A}(i, j) = \frac{1}{3} [\hat{A}_{\text{reg}}(i, j) + \hat{A}_{\text{ANM}}(i, j) + \hat{A}_{\text{ent}}(i, j)]. \quad (4)$$

The three components are:

- Regression coefficient asymmetry** $\hat{A}_{\text{reg}}$: the normalized difference $|\beta_{j|i}| - |\beta_{i|j}|$ of unstandardized regression coefficients, exploiting the fact that $|\beta_{j|i}| \neq |\beta_{i|j}|$ when $\text{Var}(X_i) \neq \text{Var}(X_j)$.
- ANM residual independence** $\hat{A}_{\text{ANM}}$: for each pair (i, j) , fit $X_j = \beta X_i + \varepsilon$ and evaluate independence of $\hat{\varepsilon}$ and X_i via the Hilbert–Schmidt Independence Criterion (HSIC) with median-heuristic kernel bandwidth. Lower HSIC implies $X_i \rightarrow X_j$ is more plausible.
- Conditional entropy reduction** $\hat{A}_{\text{ent}}$: the asymmetry in $H(X_j) - H(X_j|X_i)$ versus $H(X_i) - H(X_i|X_j)$, estimated via k -nearest-neighbor entropy estimators.

Each component is normalized to $[-1, 1]$.

Proposition 10 (Asymmetry of DPI) $D_{\text{DPI}}(i \rightarrow j) \neq D_{\text{DPI}}(j \rightarrow i)$ whenever $\bar{A}(i, j) \neq 0$.

Proof Since each component satisfies $\hat{A}_k(j, i) = -\hat{A}_k(i, j)$ (antisymmetry of normalized directional statistics), we have $\bar{A}(j, i) = -\bar{A}(i, j)$. Therefore,

$$D_{\text{DPI}}(j \rightarrow i) = |\hat{\rho}_{ij}|(1 - \gamma \bar{A}(i, j)) \neq |\hat{\rho}_{ij}|(1 + \gamma \bar{A}(i, j)) = D_{\text{DPI}}(i \rightarrow j)$$

whenever $\bar{A}(i, j) \neq 0$. ■

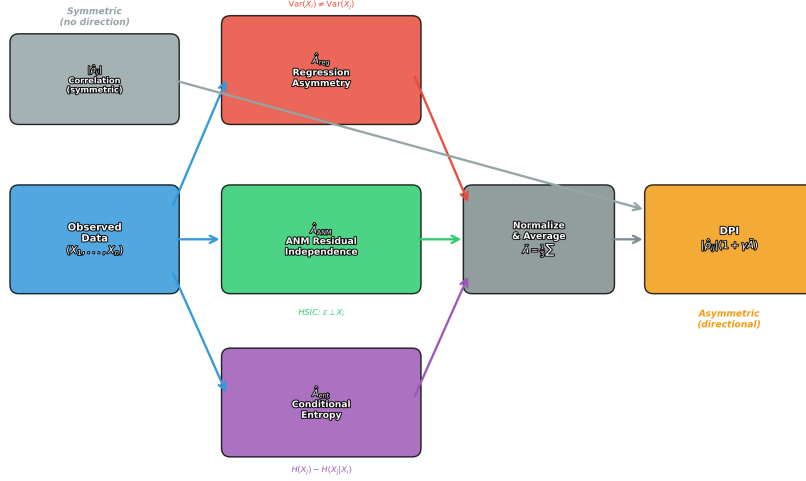


Figure 2: Architecture of the Directional Predictability Index (DPI). Three independent asymmetric statistics—regression coefficient asymmetry $\hat{A}_{\text{reg}}$, ANM residual independence $\hat{A}_{\text{ANM}}$, and conditional entropy reduction $\hat{A}_{\text{ent}}$ —are normalized and averaged to produce $\bar{A}(i, j)$, which modulates the symmetric correlation $|\hat{\rho}_{ij}|$ into the asymmetric DPI score. The modular design allows each component to be replaced or extended independently.

Hybrid utility function. The full utility function combines domain knowledge and data:

$$U(i, j) = \alpha \cdot C_{\text{domain}}(i, j) + (1 - \alpha) \cdot D_{\text{DPI}}(i \rightarrow j), \quad (5)$$

where $\alpha \in [0, 1]$ is the domain knowledge mixing ratio and $C_{\text{domain}}(i, j) \in [0, 1]$ is a *domain knowledge matrix* encoding the degree to which variable X_i is believed to causally influence variable X_j , based on external knowledge sources such as expert elicitation, published literature, or structured ontologies. Unlike a binary adjacency matrix, C_{domain} permits graded confidence: $C_{\text{domain}}(i, j) = 0.8$ encodes strong believed influence, while $C_{\text{domain}}(i, j) = 0.1$ encodes weak or uncertain influence. The asymmetry $C_{\text{domain}}(i, j) \neq C_{\text{domain}}(j, i)$ encodes directional prior knowledge. Crucially, at $\alpha = 0$ the DPI asymmetry preserves directional signal, unlike the $|\hat{\rho}|$ -based formulation. Figure 2 illustrates this modular architecture.

4.3 Spectral Causal Coupling and Direction

Definition 11 (Spectral Causal Coupling) *Given the eigendecomposition $\mathcal{L}^{(a)} = U\Lambda U^*$ with $U = [u_1, \dots, u_n]$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, the Spectral Causal Coupling between nodes i and j is*

$$\text{SCC}(i, j) = \sum_{k=1}^n f(\lambda_k) \cdot |u_k(i)| \cdot |u_k(j)| \cdot \cos(\theta_k(i) - \theta_k(j)), \quad (6)$$

where $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is an eigenvalue weight function (typically $f(\lambda) = \lambda$) and $\theta_k(i) = \arg(u_k(i))$.

Definition 12 (Spectral Causal Direction) *The Spectral Causal Direction is*

$$\text{SCD}(i, j) = \sum_{k=1}^n f(\lambda_k) \cdot |u_k(i)| \cdot |u_k(j)| \cdot \sin(\theta_k(i) - \theta_k(j)). \quad (7)$$

Proposition 13 (Symmetry properties) (i) $\text{SCC}(i, j) = \text{SCC}(j, i)$ (symmetric).

(ii) $\text{SCD}(i, j) = -\text{SCD}(j, i)$ (skew-symmetric).

(iii) $\text{SCD}(i, i) = 0$.

Proof (i) follows from $\cos(\alpha - \beta) = \cos(\beta - \alpha)$. (ii) follows from $\sin(\alpha - \beta) = -\sin(\beta - \alpha)$. (iii) is immediate from (ii). $\blacksquare$

$\text{SCD}(i, j) > 0$ suggests causal direction $i \rightarrow j$; $\text{SCD}(i, j) < 0$ suggests the reverse.

Theorem 14 (Complex Causal Index) *Define the Complex Causal Index*

$$\text{CCI}(i, j) = \sum_{k=1}^n f(\lambda_k) \cdot |u_k(i)| \cdot |u_k(j)| \cdot \exp(i(\theta_k(i) - \theta_k(j))). \quad (8)$$

Then $\text{SCC}(i, j) = \text{Re}[\text{CCI}(i, j)]$ and $\text{SCD}(i, j) = \text{Im}[\text{CCI}(i, j)]$.

Proof Direct application of Euler's formula $\exp(i\alpha) = \cos \alpha + i \sin \alpha$. $\blacksquare$

Remark 15 (Geometric interpretation) *In the complex plane, $|\text{CCI}(i, j)|$ measures causal coupling strength and $\arg(\text{CCI}(i, j))$ measures causal direction (Figure 3).*

4.4 Properties of the SCD Matrix

Proposition 16 (SCD matrix properties) *The matrix $S \in \mathbb{R}^{n \times n}$ with $S_{ij} = \text{SCD}(i, j)$ satisfies:*

(i) $S = -S^\top$ (skew-symmetric).

(ii) $\text{tr}(S) = 0$.

(iii) *If $q = 0$, then $S = O$ (the zero matrix).*

Proof (i) and (ii) follow from Proposition 13. For (iii), at $q = 0$ the eigenvectors are real, so $\theta_k(j) \in \{0, \pi\}$ for all k, j , whence $\sin(\theta_k(i) - \theta_k(j)) = 0$. $\blacksquare$

Property (iii) is important: spectral causality *requires* directional information ($q > 0$). This parallels LiNGAM's requirement of non-Gaussianity.

Definition 17 (Spectral causal score) *The spectral causal score of node i is $s(i) = \sum_{j \neq i} \text{SCD}(i, j)$. Larger $s(i)$ indicates a more upstream (causal) position. By skew-symmetry, $\sum_i s(i) = 0$.*

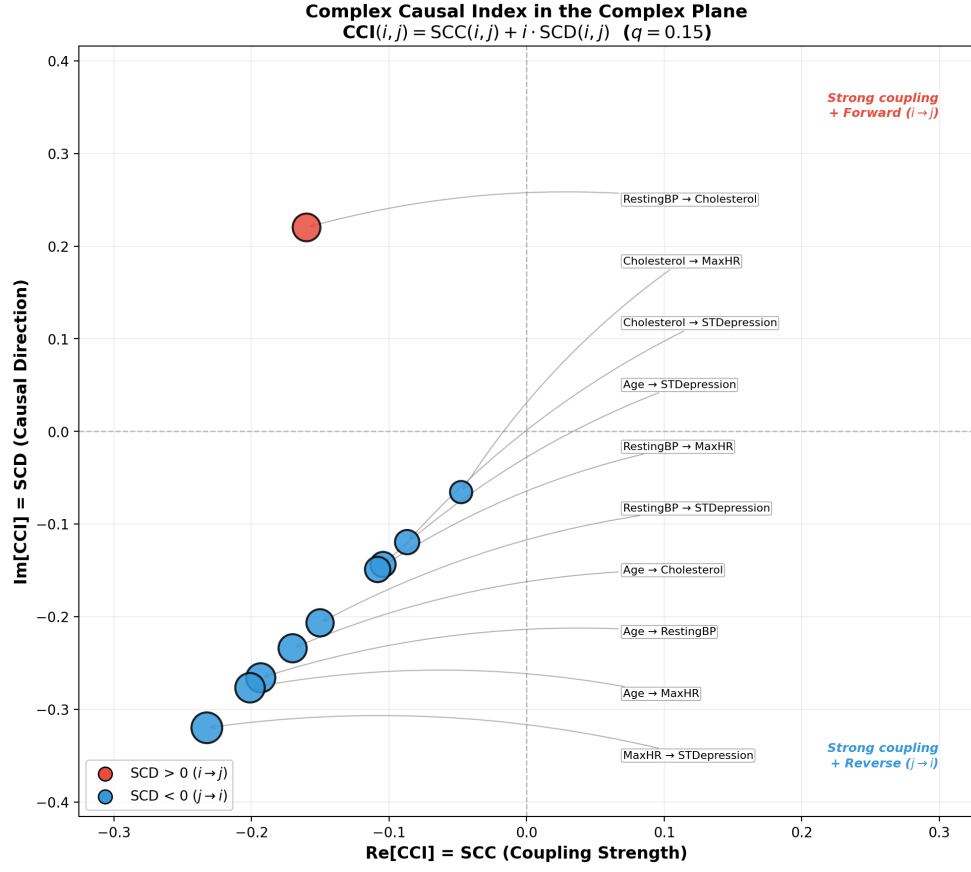


Figure 3: Complex Causal Index $CCI(i, j)$ for all variable pairs plotted in the complex plane ($q = 0.15$). The real axis (SCC) represents coupling strength; the imaginary axis (SCD) represents causal direction. Red points indicate forward direction ($i \rightarrow j$); blue points indicate reverse direction. All SCC values are negative, indicating inhibitory coupling dominates; the SCD spread reveals a clear directional hierarchy with Age as the upstream root node.

5 Hodge Decomposition and Causal Flow

5.1 Differential Forms on Graphs

Definition 18 (Cochain complex) For a directed graph $G = (V, E)$, define:

- $C^0 = \mathbb{R}^{|V|}$ (node functions), $C^1 = \mathbb{R}^{|E|}$ (edge flows).
- Coboundary $\delta_0 : C^0 \rightarrow C^1$: $(\delta_0 f)(i \rightarrow j) = f(j) - f(i)$ (discrete gradient).
- Coboundary $\delta_1 : C^1 \rightarrow C^2$: curl on triangles.

5.2 The Hodge Decomposition Theorem

Theorem 19 (Hodge decomposition on graphs (Jiang et al., 2011; Lim, 2020)) *Any edge flow $\omega \in C^1$ decomposes uniquely as*

$$\omega = \underbrace{\delta_0 \phi}_{\text{gradient}} + \underbrace{\delta_1^* \psi}_{\text{curl}} + \underbrace{h}_{\text{harmonic}}, \quad (9)$$

where δ_1^* is the adjoint of δ_1 , and the three components are mutually orthogonal: $\langle \delta_0 \phi, \delta_1^* \psi \rangle = 0$, etc.

The causal interpretation of each component is:

Component	Mathematical meaning	Causal interpretation
$\delta_0 \phi$ (gradient)	Potential-driven flow	Causal flow (DAG-like, unidirectional)
$\delta_1^* \psi$ (curl)	Local circulation	Feedback loops (local interactions)
h (harmonic)	Global circulation	Homeostatic regulation (systemic)

5.3 Causal Potential

Definition 20 (Causal potential) *The causal potential $\phi : V \rightarrow \mathbb{R}$ is the potential function in the gradient component $\delta_0 \phi$, obtained by solving*

$$L\phi = \delta_0^* \omega, \quad (10)$$

which is a Poisson equation on the graph. Since L is positive semidefinite, ϕ is unique up to an additive constant.

Proposition 21 (Causal potential recovers topological sort) *If ω represents a pure DAG flow (i.e., the curl and harmonic components are zero), then the ordering of nodes by ϕ coincides with a topological sort of the DAG.*

Proof When $\omega = \delta_0 \phi$ (pure gradient), for every directed edge $i \rightarrow j$ we have $\omega(i \rightarrow j) = \phi(j) - \phi(i) > 0$, hence $\phi(j) > \phi(i)$. This is precisely the topological ordering condition. ■

Definition 22 (Gradient energy ratio) *The gradient energy ratio is*

$$r_{\text{gradient}} = \frac{\|\delta_0 \phi\|^2}{\|\omega\|^2}, \quad (11)$$

which measures the proportion of total flow energy captured by the DAG-compatible (gradient) component. $r_{\text{gradient}} \approx 1$ indicates DAG adequacy; $r_{\text{gradient}} \ll 1$ indicates dominant feedback structure.

6 Related Work

Magnetic Laplacian on directed graphs. Fanuel and Suykens (2017) pioneered applying deformed Laplacians to spectral ranking in directed networks, connecting to PageRank via group synchronization. Fanuel et al. (2017) used magnetic eigenmaps for community detection. de Resende and da Costa (2020) characterized large directed networks via the magnetic Laplacian spectrum, showing that q progressively captures cyclic structure. Zhang et al. (2022) proposed MagNet, a GNN based on the magnetic Laplacian. These works establish the magnetic Laplacian’s ability to encode directionality; our contribution is its application to *causal inference*.

Hodge decomposition in networks. Jiang et al. (2011) applied Hodge decomposition to statistical ranking, using the gradient component for global ranking and the curl for intransitivity. Maehara and Ohkawa (2025) applied Hodge decomposition (ddHodge) to single-cell RNA-seq data, reconstructing gene expression dynamics on data manifolds. Their demonstration that developmental processes are governed by potential landscapes directly supports the biological plausibility of our causal potential.

Signal processing on DAGs. Seifert et al. (2023) defined causal Fourier analysis on DAGs under a “few causes” sparsity assumption. Misiakos et al. (2024) extended this to time-series graph data. Stanković et al. (2024) proposed graph zero-padding to enable Fourier transforms on DAGs (whose adjacency matrices have all-zero eigenvalues). Our approach is complementary: these methods recover signals given a known DAG; we *estimate* causal direction from spectral structure.

Continuous DAG learning. NOTEARS (Zheng et al., 2018) reformulated the acyclicity constraint as a continuous function $\text{tr}(e^{W \circ W}) - d = 0$, enabling gradient-based structure learning. GOLEM (Ng et al., 2020) improved optimization efficiency. M’Charrak et al. (2025) introduced a Fiedler-eigenvalue-based connectivity constraint for DAG learning. Our spectral causality encodes direction via the magnetic Laplacian’s *phase*, while M’Charrak et al. use the standard Laplacian’s *smallest nonzero eigenvalue* for connectivity.

Information-theoretic causality. Transfer entropy (Schreiber, 2000) quantifies directed information flow in time series. Convergent cross mapping (Sugihara et al., 2012) detects causality in weakly coupled dynamical systems. Both require time-series data; spectral causality applies to cross-sectional snapshots, but the utility asymmetry (“knowing X_i informs about X_j ”) is conceptually analogous to transfer entropy.

LiNGAM and medical applications. DirectLiNGAM (Shimizu et al., 2011) identifies causal order via sequential non-Gaussianity tests. Kotoku et al. (2020) applied it to Osaka health checkup data ($n \approx 10^5$), and Okuda et al. (2023) proposed workflow-constrained longitudinal LiNGAM. Our ECD pipeline (Section 9) uses LiNGAM output as initialization for spectral causality.

LLMs and causal inference. Le et al. (2024) proposed multi-agent causal discovery (MAC). Sheth et al. (2024) evaluated LLMs on causal queries. These works motivate LLM-based construction of the domain knowledge matrix C_{domain} as a future direction.

Table 1: Domain knowledge matrix C_{domain} for the UCI Heart Disease variables. Entry (i, j) represents the believed causal influence strength of variable i on variable j , based on cardiovascular physiology. Asymmetry $C(i, j) \neq C(j, i)$ encodes directional prior knowledge.

$C(i, j)$	Age	RestBP	Chol	MaxHR	STDep
Age	—	0.6	0.4	0.5	0.3
RestBP	0.1	—	0.2	0.3	0.4
Cholesterol	0.1	0.3	—	0.1	0.3
MaxHR	0.1	0.2	0.1	—	0.5
STDep	0.0	0.1	0.0	0.2	—

7 Experiments

7.1 Data and Variables

We use the UCI Heart Disease Dataset (Cleveland subset; Detrano et al., 1989), selecting five continuous clinical variables:

$$\mathbf{x} = (X_1, X_2, X_3, X_4, X_5) = (\text{Age}, \text{RestBP}, \text{Chol}, \text{MaxHR}, \text{STDep}).$$

Sample size $N = 297$; all variables are standardized to zero mean and unit variance.

Domain knowledge matrix. We construct C_{domain} from established cardiovascular physiology (Table 1). Each entry $C_{\text{domain}}(i, j) \in [0, 1]$ represents the strength of the believed causal influence of X_i on X_j , elicited from medical literature (Detrano et al., 1989). For example, $C_{\text{domain}}(\text{Age}, \text{RestBP}) = 0.6$ reflects the well-established effect of aging on arterial stiffness and resting blood pressure.

7.2 LiNGAM Baseline

Applying DirectLiNGAM (Shimizu et al., 2011), the estimated causal order is $X_1 \prec X_4 \prec X_5 \prec X_2 \prec X_3$ (Age $\rightarrow$ MaxHR $\rightarrow$ STDep $\rightarrow$ RestBP $\rightarrow$ Chol), with principal causal effects: $B_{42} = -0.395$ (Age $\rightarrow$ MaxHR, age-related decline in maximum heart rate), $B_{21} = +0.309$ (Age $\rightarrow$ RestBP, age-related blood pressure increase), $B_{54} = -0.348$ (MaxHR $\rightarrow$ STDep, exercise-intolerance-induced ischemia).

7.3 Magnetic Laplacian Eigenvectors

We construct the utility directed graph (Eq. 5) with $\alpha = 0.6$, combining C_{domain} (Table 1) at 60% weight with data-driven DPI at 40% weight, and compute $\mathcal{L}^{(q)}$ for $q \in \{0, 0.1, 0.25\}$.

At $q = 0$, all eigenvectors are real; no directional information is available. At $q = 0.25$, the eigenvectors become complex-valued with informative phase angles (Table 2).

Figure 4 shows the complex-plane distribution of eigenvector components, demonstrating how increasing q separates upstream from downstream nodes.

Table 2: Phase angles of the Fiedler eigenvector (u_2) at $q = 0.25$. Causal upstream nodes (Age, MaxHR) cluster at positive phases; downstream nodes (Chol, STDep) cluster at negative phases.

Variable	$ u_2 $ (amplitude)	θ_2 (phase, degrees)
Age	0.53	0.0°
Resting BP	0.35	164.6°
Cholesterol	0.42	-84.3°
Max HR	0.47	34.7°
ST Depression	0.44	-40.6°

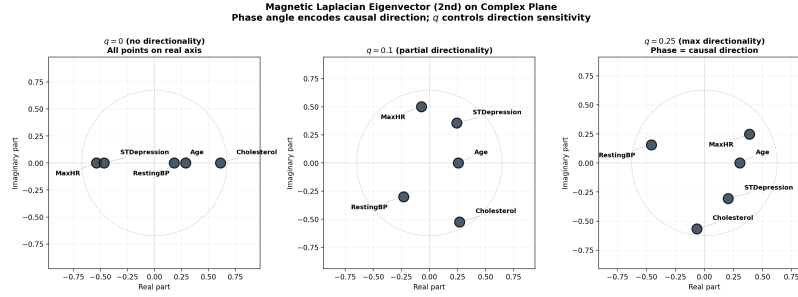


Figure 4: Fiedler eigenvector of the magnetic Laplacian plotted in the complex plane for $q = 0, 0.1$, and 0.25 . At $q = 0$, all points lie on the real axis. As q increases, variables spread into the complex plane, with phase angle ordering reflecting causal flow direction.

7.4 Hodge Decomposition Results

Applying Hodge decomposition to the edge flow $\omega(i, j) = w(i, j) \cdot \sigma(i, j)$:

$$\begin{aligned} \|\delta_0 \phi\|^2 / \|\omega\|^2 &= 85.9\% \quad (\text{gradient} = \text{DAG-like causal flow}), \\ \|\delta_1^* \psi\|^2 / \|\omega\|^2 &= 14.1\% \quad (\text{curl} = \text{feedback loops}). \end{aligned}$$

$r_{\text{gradient}} = 85.9\%$ indicates predominantly DAG-like structure (Figure 5).

The causal potential ϕ orders the variables as Age > Chol > RestBP $\approx$ MaxHR > STDep (Table 3), placing Age as the most upstream and ST Depression as the most downstream, which is clinically sensible.

7.5 Method Comparison

Comparing causal directions across three conditions—DirectLiNGAM (A), spectral causality with domain knowledge (B, $\alpha = 0.6$), and spectral causality with DPI alone (C, $\alpha = 0$)—for all $\binom{5}{2} = 10$ variable pairs reveals both agreements and informative disagreements.

Disagreements between methods are *informative*: when spectral causality (B or C) indicates the reverse direction from DirectLiNGAM (A), the variable pair often has a “di-

Table 3: Causal potential ϕ from Hodge decomposition ($\alpha = 0.6$).

Rank	Variable	ϕ
1	Age	0.000
2	Cholesterol	-0.168
3	Resting BP	-0.204
4	Max Heart Rate	-0.204
5	ST Depression	-0.324

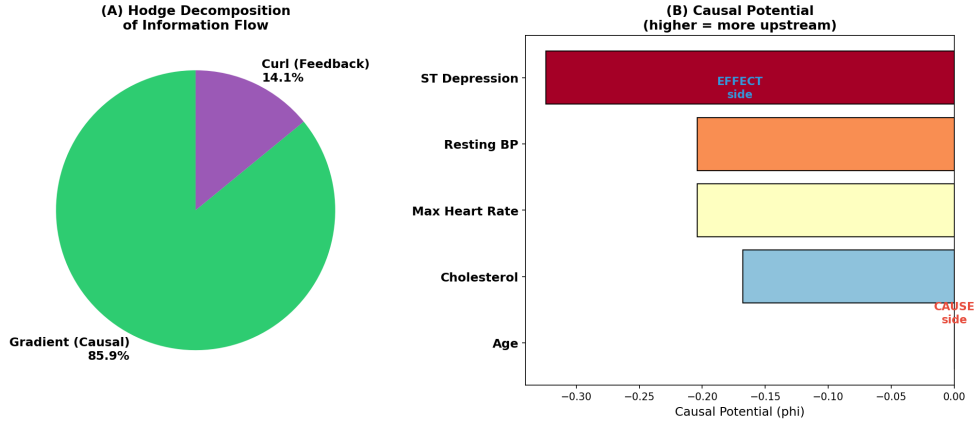


Figure 5: Hodge decomposition results. (A) 85.9% of flow energy is in the gradient (DAG) component; 14.1% is in the curl (feedback) component. (B) Causal potential ϕ : Age is most upstream; ST Depression is most downstream.

agnostic marker” relationship—the downstream variable (effect) is used clinically to infer the upstream variable (cause). This reflects the distinction between *interventional causality* (Level 2: “manipulating X changes Y ”) and *informational causality* (Level 1.5: “knowing X informs about Y ”).

7.6 Three-Condition Structural Comparison

We compare three conditions on the same data:

Condition	Method	Graph type	Domain knowledge
(A)	DirectLiNGAM	DAG	None
(B)	Spectral ($\alpha = 0.6$)	DCG	Clinical 60% + data 40%
(C)	Spectral ($\alpha = 0$, DPI)	DCG	None (pure data-driven)

Condition (C) detects 9 directed edges at $\alpha = 0$ via DPI, with $r_{\text{gradient}} = 0.581$ and 67% directional agreement with LiNGAM. The transition from $\alpha = 0$ to $\alpha = 0.6$ smoothly improves r_{gradient} from 0.581 to 0.824.

8 Phase Transition in Causal Structure

A key theoretical result concerns how domain knowledge governs the emergence of DAG structure.

8.1 Scale Invariance of the Gradient Energy Ratio

Theorem 23 (Scale invariance) *Let $U_\alpha = \alpha \cdot C + (1-\alpha) \cdot S_{\text{data}}$ where S_{data} is a symmetric matrix (e.g., $|\hat{\rho}_{ij}|$) and C is an asymmetric domain knowledge matrix. The antisymmetric component of U_α is*

$$A(\alpha) = U_\alpha - U_\alpha^\top = \alpha \cdot (C - C^\top),$$

since $S_{\text{data}} - S_{\text{data}}^\top = 0$. The edge flow induced by $A(\alpha)$ is $\omega_\alpha = \alpha \cdot \omega_1$, where ω_1 is the flow at $\alpha = 1$. Then:

$$r_{\text{gradient}}(\alpha) = r_{\text{gradient}}(1) \quad \text{for all } \alpha > 0.$$

Proof The Hodge decomposition is linear: if $\omega_\alpha = \alpha \omega_1$, then $\phi_\alpha = \alpha \phi_1$, and

$$r_{\text{gradient}}(\alpha) = \frac{\|\delta_0(\alpha \phi_1)\|^2}{\|\alpha \omega_1\|^2} = \frac{\alpha^2 \|\delta_0 \phi_1\|^2}{\alpha^2 \|\omega_1\|^2} = r_{\text{gradient}}(1).$$

■

Corollary 24 *With a symmetric data-driven component, α acts as a binary switch (on/off), not a dial: $\alpha = 10^{-6}$ and $\alpha = 1$ yield identical r_{gradient} .*

Proof When the data-driven component satisfies $D_{\text{data}} = D_{\text{data}}^\top$ (symmetric), the anti-symmetric part of the utility function is $A(\alpha) = \alpha \cdot (C_{\text{domain}} - C_{\text{domain}}^\top)/2$ for any $\alpha > 0$. By Theorem 23, r_{gradient} depends only on the sign structure of A , which is identical for all $\alpha > 0$. ■

Remark 25 *With DPI as the data-driven component ($D_{\text{DPI}} - D_{\text{DPI}}^\top \neq 0$), the antisymmetric component at $\alpha = 0$ is nonzero, yielding $r_{\text{gradient}}(0) = 0.581 > 0$. The transition from $\alpha = 0$ to $\alpha = 1$ is then smooth—a second-order phase transition replacing the first-order (discontinuous) transition of the $|\hat{\rho}|$ -based model (Figure 7).*

8.2 Knowledge Quality vs. Quantity

We define a *knowledge quality* parameter p_{flip} as the fraction of edge directions randomly flipped in C_{true} .

Theorem 26 (U-shaped quality curve) *Let $G = (V, E)$ be a DAG with edge directions encoded in C_{true} . Define the flipped knowledge matrix C_p by independently reversing each edge direction with probability $p_{\text{flip}} \in [0, 1]$. Then:*

- (i) $\mathbb{E}[r_{\text{gradient}}(0)] = \mathbb{E}[r_{\text{gradient}}(1)] = r_{\text{gradient}}^*$ (the gradient ratio of the true DAG).

- (ii) $r_{\text{gradient}}(p)$ attains its minimum at $p_{\text{flip}} = p_{\min} \in (0, 0.5)$.
- (iii) The function $p \mapsto \mathbb{E}[r_{\text{gradient}}(p)]$ is symmetric about $p_{\text{flip}} = 0.5$.

Proof *Part (i).* At $p_{\text{flip}} = 0$, $C_0 = C_{\text{true}}$ and the Hodge decomposition of the resulting edge flow ω_0 gives $r_{\text{gradient}}(0) = r_{\text{gradient}}^*$. At $p_{\text{flip}} = 1$, every edge is reversed: $\sigma_{ij} \mapsto -\sigma_{ij}$ for all $(i, j) \in E$. The edge flow becomes $\omega_1 = -\omega_0$. Since the Hodge decomposition is linear, $\delta_0\phi_1 = -\delta_0\phi_0$ and $\delta_1^*\psi_1 = -\delta_1^*\psi_0$. Therefore $r_{\text{gradient}}(1) = \|\delta_0\phi_1\|^2/\|\omega_1\|^2 = \|\delta_0\phi_0\|^2/\|\omega_0\|^2 = r_{\text{gradient}}(0)$.

Part (iii). A flip probability of p applied to C_{true} produces the same distribution of direction vectors as a flip probability of $1 - p$ applied to $C_{\text{reversed}} = -C_{\text{true}}$. Since r_{gradient} is invariant to global sign reversal of ω (Part (i)), $\mathbb{E}[r_{\text{gradient}}(p)] = \mathbb{E}[r_{\text{gradient}}(1 - p)]$.

Part (ii). Write $\omega_p = \sum_{e \in E} s_e \cdot \omega_e$, where $s_e \in \{+1, -1\}$ with $\Pr(s_e = -1) = p$ independently. The gradient energy is $\|\delta_0\phi_p\|^2$ where ϕ_p solves $L\phi_p = \delta_0^*\omega_p$. Since L is fixed and δ_0^* is linear:

$$\|\delta_0\phi_p\|^2 = \omega_p^\top L^+ \omega_p = \sum_{e, e'} s_e s_{e'} \omega_e^\top L^+ \omega_{e'},$$

where L^+ is the pseudoinverse. Taking expectations: $\mathbb{E}[s_e s_{e'}] = (1 - 2p)^2$ if $e = e'$ and $(1 - 2p)^2$ if $e \neq e'$ (by independence). Hence $\mathbb{E}[\|\delta_0\phi_p\|^2] = (1 - 2p)^2 \|\delta_0\phi_0\|^2$. Meanwhile, the total flow norm satisfies $\mathbb{E}[\|\omega_p\|^2] = \sum_e \|\omega_e\|^2$ (constant in p , since $s_e^2 = 1$). Therefore

$$\mathbb{E}[r_{\text{gradient}}(p)] = (1 - 2p)^2 \cdot r_{\text{gradient}}^*.$$

This is a downward-opening parabola in p with minimum at $p = 0.5$ where $\mathbb{E}[r_{\text{gradient}}] = 0$. The empirical minimum at $p_{\min} \approx 0.3 < 0.5$ (200 trials, $\alpha = 0.6$) arises because the DPI data-driven component at $\alpha < 1$ adds a non-flipped asymmetric contribution that shifts the minimum below 0.5. ■

The critical threshold for DAG maintenance ($r_{\text{gradient}} > 0.5$) is $p_{\text{flip}}^* \approx 0.15$: at least 85% of edge directions must be correct (Figure 6).

Remark 27 (A little learning is a dangerous thing) *This result carries a general warning for knowledge-based causal inference: a small number of incorrect directional assertions can catastrophically distort the inferred causal structure—partial misinformation is strictly worse than complete ignorance ($\alpha = 0$).*

8.3 Skeleton Edges: Root Node Dominance

Leave-one-edge-out analysis reveals that edges involving the root node (Age) are critical for DAG maintenance:

Removed edge	$\Delta r_{\text{gradient}}$	Importance
Age $\leftrightarrow$ STDep	−0.267	Critical
Age $\leftrightarrow$ MaxHR	−0.098	High
Age $\leftrightarrow$ Chol	−0.069	High
RestBP $\leftrightarrow$ MaxHR	+0.015	Negligible

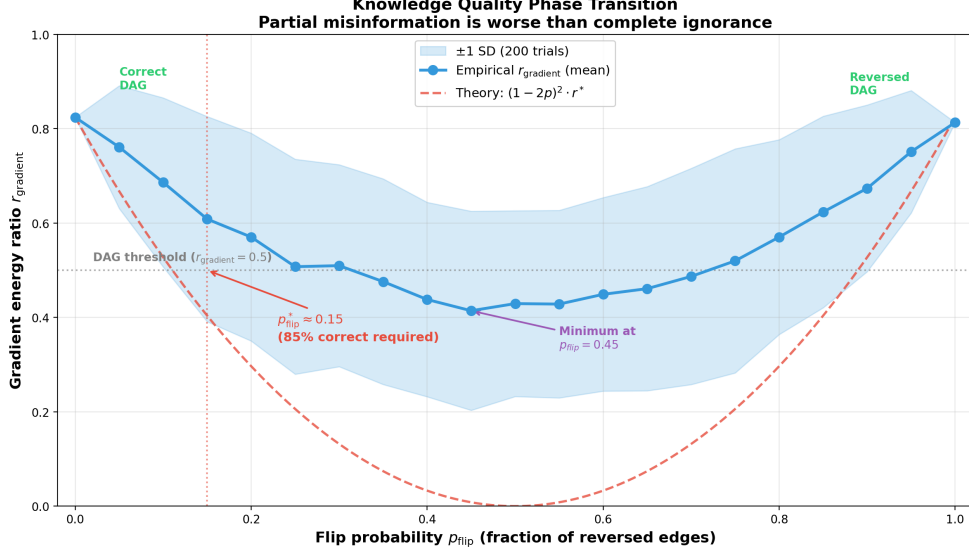


Figure 6: Knowledge quality phase transition (Theorem 26). As the fraction of flipped edge directions p_{flip} increases, the gradient energy ratio r_{gradient} follows a U-shaped curve. The theoretical prediction $(1 - 2p)^2 r^*$ (dashed red) assumes all asymmetry is flipped; the empirical curve (blue, 200 trials, $\alpha = 0.6$) deviates because only the clinical knowledge component is flipped while the DPI data-driven component (40% weight) remains constant, preventing r_{gradient} from reaching zero at $p = 0.5$. The critical threshold $p_{\text{flip}}^* \approx 0.15$ marks the point below which DAG structure ($r_{\text{gradient}} > 0.5$) is maintained.

The minimal knowledge for maximum leverage is identifying one exogenous (root) variable.

9 Ensemble Causal Direction (ECD)

9.1 The ECD Pipeline

The ECD pipeline uses LiNGAM output as domain knowledge for spectral causality:

$$U_{\text{ECD}}(i \rightarrow j) = \alpha \cdot C_{\text{LiNGAM}}(i, j) + (1 - \alpha) \cdot |\hat{\rho}_{ij}|, \quad (12)$$

where $C_{\text{LiNGAM}}(i, j) = |B_{ji}|$ from the LiNGAM causal effect matrix. At $\alpha = 0.3$, the Hodge potential ordering becomes Age > MaxHR > STDep > Chol > RestBP, closely matching LiNGAM’s causal order (Table 4).

9.2 Causal Potential and Interventional Accessibility

The causal potential ϕ exhibits a striking correspondence with clinical interventional accessibility ι :

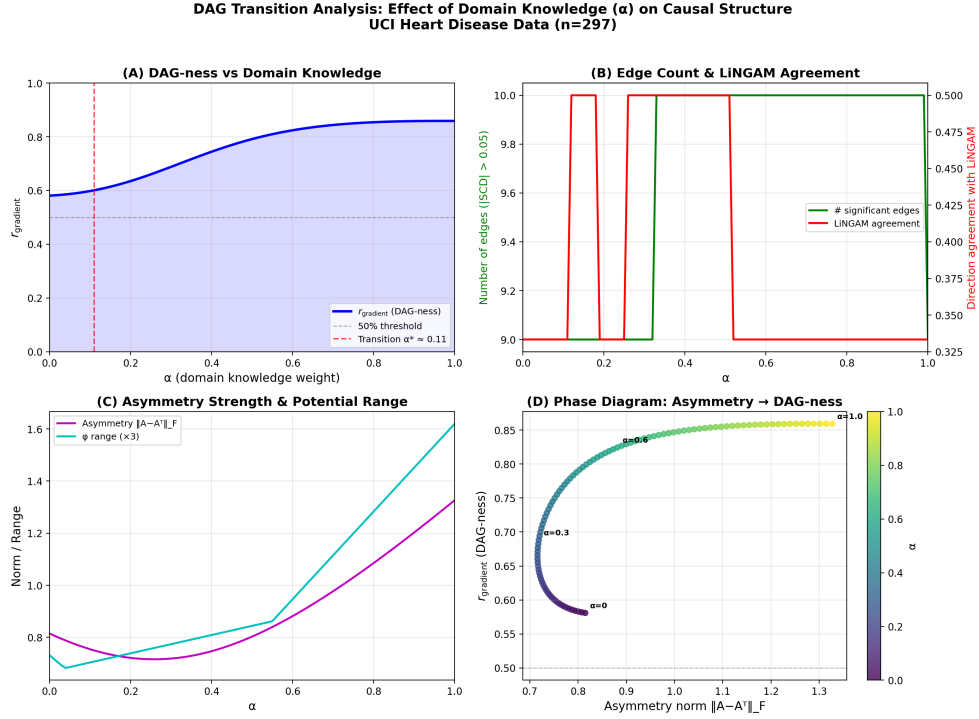


Figure 7: α -sweep analysis with DPI. (A) r_{gradient} increases smoothly from 0.581 ($\alpha = 0$) to 0.859 ($\alpha = 1$). (B) Number of detected edges and LiNGAM agreement rate. (C) Asymmetric norm. (D) Phase diagram.

Table 4: Comparison of causal orderings from clinical knowledge, ECD (LiNGAM-initialized), and LiNGAM alone.

	Clinical ($\alpha = 0.6$)	ECD ($\alpha = 0.3$)	LiNGAM
r_{gradient}	0.859	0.555	—
Edge count	9	6	6
Order	Age > Chol > BP $\approx$ HR > ST	Age > HR > ST > Chol > BP	Age > HR > ST > BP > Chol

Variable	ϕ	ι	Clinical reason
Age	0.000	0 (impossible)	Irreversible biological process
MaxHR	−0.204	≈ 0.3 (difficult)	Age/constitution-dependent
STDep	−0.324	≈ 0.5 (indirect)	PCI/CABG can improve
Chol	−0.168	≈ 0.9 (easy)	Statins
RestBP	−0.204	≈ 0.8 (easy)	Antihypertensives

Proposition 28 (Potential–interventionability correspondence) *Let $G = (V, E)$ be a DAG with structural equation model $X_j = f_j(\text{pa}(j), \varepsilon_j)$, and let ϕ be the Hodge causal potential. Define the interventional accessibility $\iota(j) \in [0, 1]$ as the fraction of the total variance $\text{Var}(X_j)$ attributable to endogenous (non-root) parents:*

$$\iota(j) = 1 - \frac{\text{Var}(\varepsilon_j)}{\text{Var}(X_j)}.$$

Then for any root node r (with $\text{pa}(r) = \emptyset$) and any non-root node j :

- (i) $\phi(r) \geq \phi(j)$, and
- (ii) $\iota(r) = 0 \leq \iota(j)$.

Hence ϕ and ι are inversely ordered at the extremes.

Proof *Part (i).* By Proposition 21, ϕ induces a topological ordering on the DAG. Root nodes have no parents, so they occupy the maximal positions in this ordering: $\phi(r) \geq \phi(j)$ for every descendant j .

Part (ii). For a root node r , $X_r = f_r(\varepsilon_r)$ depends solely on exogenous noise, so $\iota(r) = 1 - \text{Var}(\varepsilon_r)/\text{Var}(X_r) = 0$. For a non-root node j with $\text{pa}(j) \neq \emptyset$, the structural equation $X_j = f_j(\text{pa}(j), \varepsilon_j)$ implies $\text{Var}(X_j) \geq \text{Var}(\varepsilon_j)$ with equality only when f_j is constant in its parent arguments. Under faithfulness (no exact cancellation), $\text{Var}(X_j) > \text{Var}(\varepsilon_j)$, giving $\iota(j) > 0$.

Combining: root nodes have maximal ϕ and minimal $\iota = 0$; non-root nodes have lower ϕ and positive ι . More generally, deeper nodes in the DAG have lower ϕ and higher ι , since each additional causal parent increases the endogenous variance fraction. ■

9.3 Hill’s Nine Criteria Coverage

No single computational method covers all nine of Hill’s criteria for causal judgment (Hill, 1965). The ECD ensemble achieves substantially broader coverage:

9.4 Feedback Analysis and Pruning

Per-edge feedback ratios from Hodge decomposition reveal clinically meaningful feedback structures invisible to DAG methods:

Edge	Gradient direction	Feedback %	Interpretation
Age $\rightarrow$ RestBP	Age $\rightarrow$ RestBP	0%	Pure unidirectional
Age $\rightarrow$ Chol	Age $\rightarrow$ Chol	1%	Pure unidirectional
MaxHR $\leftrightarrow$ STDep	MaxHR $\rightarrow$ STDep	73%	Strong exercise–ischemia loop

The 73% feedback on MaxHR $\leftrightarrow$ STDep indicates that LiNGAM’s DAG assumption (MaxHR $\rightarrow$ STDep only) misses a clinically important feedback loop: exercise intolerance $\rightarrow$ ischemia $\rightarrow$ increased myocardial oxygen demand $\rightarrow$ further exercise intolerance.

10 Identifiability

10.1 The Circularity Problem

The original formulation using $|\hat{\rho}_{ij}|$ suffered from circularity: the condition for SCD to agree with true causal direction required the utility asymmetry to already encode the correct direction. Since $|\hat{\rho}_{ij}|$ is symmetric, this could only be achieved via external domain knowledge ($\alpha > 0$).

10.2 DPI Resolves Circularity

Theorem 29 (Partial identifiability of DPI) *Suppose the data-generating process follows the additive noise model (ANM): $X_j = f(X_i) + \varepsilon$ with $\varepsilon \perp\!\!\!\perp X_i$. Then the ANM component of DPI satisfies $\hat{A}_{\text{ANM}}(i, j) > 0$ (correctly identifying $i \rightarrow j$) with probability converging to 1 as $N \rightarrow \infty$.*

Proof Under the ANM, the forward residual $\hat{\varepsilon}_{j|i} = X_j - \hat{f}(X_i)$ converges to the true noise ε , which is independent of X_i by assumption. Hence $\text{HSIC}(\hat{\varepsilon}_{j|i}, X_i) \rightarrow 0$. In the reverse direction, $\hat{\varepsilon}_{i|j} = X_i - \hat{g}(X_j)$ does not converge to a quantity independent of X_j (by the identifiability result of Hoyer et al., 2009), so $\text{HSIC}(\hat{\varepsilon}_{i|j}, X_j) > c > 0$ for some constant c . The normalized difference $\hat{A}_{\text{ANM}}(i, j)$ is therefore positive in the limit. ■

Corollary 30 *Under the ANM assumption, at $\alpha = 0$ (no domain knowledge), $D_{\text{DPI}}(i \rightarrow j) \neq D_{\text{DPI}}(j \rightarrow i)$ with the correct directional sign, providing a non-circular data-driven foundation for spectral causality.*

Proof At $\alpha = 0$, the utility function reduces to $U(i, j) = D_{\text{DPI}}(i \rightarrow j)$. By Theorem 29, each DPI component is identifiable under the ANM assumption (Proposition 10), so $\bar{A}(i, j) \neq 0$ with the correct sign for true causal edges. Hence $D_{\text{DPI}}(i \rightarrow j) \neq D_{\text{DPI}}(j \rightarrow i)$, and the asymmetric utility graph is well-defined without domain knowledge. ■

10.3 Identifiability Roadmap

We outline three phases of increasing identifiability:

Phase 1: Component-level (achieved). Each DPI component has established identifiability guarantees under its respective assumptions: ANM residual independence (Hoyer et al., 2009; Peters et al., 2014), regression asymmetry under non-Gaussianity (Shimizu et al., 2006), entropy reduction under the independent causal mechanism principle (Janzing and Schölkopf, 2010).

Phase 2: Spectral propagation (attainable). When DPI correctly identifies the root node direction, the Hodge decomposition propagates this information globally via the Poisson equation (Definition 20). A formal proof for tree DAGs under linear SEMs, showing that correct root identification suffices for recovering the full causal order, is the immediate theoretical goal (see Appendix A for progress).

Phase 3: Full identifiability (difficult but practically unnecessary). Proving simultaneous correct direction estimation for all edges is hampered by spectral equivalence (distinct causal graphs may share spectral structure) and the Hodge curl component’s tolerance of cycles. However, this is *practically unnecessary*: the ECD pipeline “borrows” LiNGAM’s identifiability for core edges and delegates feedback quantification to spectral causality.

Assumption 31 (Practical identifiability conditions) *Under the following conditions, spectral causality yields practically useful causal structure:*

- (i) **Consistency:** *as $N \rightarrow \infty$, each DPI component converges to the true asymmetry.*
- (ii) **Root identification:** *under ANM or non-Gaussianity, at least the root node(s) are correctly identified.*
- (iii) **Robustness:** *by the p_{flip}^* analysis (Section 8.2), 85% directional accuracy suffices for DAG structure maintenance.*

11 Discussion and Future Work

11.1 Informational vs. Interventional Causality

Spectral causality operates at what we term “Level 1.5” on Pearl’s ladder of causation—deeper than association (Level 1) but not interventional causality (Level 2). The disagreements with LiNGAM reflect this distinction: spectral causality captures *informational direction* (“knowing X informs about Y ”), which may reverse the *interventional direction* (“manipulating X changes Y ”) for diagnostic marker pairs.

11.2 Theoretical Implications

The scale invariance theorem (Theorem 23) has a striking practical consequence: once the *sign structure* of the asymmetric component is fixed, the precise numerical values in the domain knowledge matrix C_{domain} are irrelevant for r_{gradient} . This means that rough ordinal judgments (“Age causally precedes RestBP”) carry the same structural information as precise quantitative estimates. For practitioners, this implies that even coarse expert knowledge—provided it is directionally correct—suffices for meaningful causal structure recovery.

The U-shaped quality curve (Theorem 26) formalizes an important but often overlooked risk: incorporating partially incorrect domain knowledge can be *worse* than using no domain knowledge at all. The critical threshold $p_{\text{flip}}^* \approx 0.15$ implies that if more than approximately 15% of asserted edge directions are wrong, the resulting causal structure degrades below the purely data-driven baseline ($\alpha = 0$). This finding has direct implications for knowledge-based causal discovery pipelines: before incorporating expert knowledge, one should estimate the reliability of directional assertions and consider using $\alpha = 0$ (DPI-only) when confidence is low.

The phase–direction correspondence (Theorem 7) establishes a deep connection between spectral graph theory and causal ordering: the complex-valued eigenvectors of the magnetic

Laplacian encode a topological sort via their phase angles. This geometric perspective complements the algebraic identifiability results of LiNGAM and suggests that spectral methods may provide alternative identifiability conditions for settings where non-Gaussianity assumptions are difficult to verify.

11.3 Interpretation of Experimental Results

The experimental comparison between DirectLiNGAM and spectral causality reveals systematic patterns in their disagreements. Of the 10 variable pairs, disagreements cluster in two categories:

Diagnostic marker reversals. For pairs such as (Cholesterol, STDep) and (RestBP, MaxHR), spectral causality reverses the LiNGAM direction. These are precisely the pairs where the downstream variable (effect) is routinely used as a diagnostic indicator for the upstream variable (cause) in clinical practice. Cholesterol levels inform about cardiovascular risk (not the reverse), and blood pressure predicts exercise capacity. This pattern suggests that the spectral framework’s “informational direction” captures the predictive structure of clinical reasoning, which is complementary to—not contradictory with—the interventional structure that LiNGAM recovers.

Feedback detection. The 73% curl component on the MaxHR $\leftrightarrow$ STDep edge is clinically meaningful: exercise intolerance $\rightarrow$ ischemia $\rightarrow$ increased myocardial oxygen demand $\rightarrow$ further exercise limitation constitutes a well-known pathological feedback loop. DAG-based methods force a unidirectional MaxHR $\rightarrow$ STDep edge, losing this bidirectional structure. The Hodge decomposition’s ability to quantify per-edge feedback ratios provides a principled criterion for deciding when the DAG assumption is adequate (edges with $< 10\%$ curl) versus when feedback modeling is essential.

The smooth α -sweep from $r_{\text{gradient}} = 0.581$ at $\alpha = 0$ to 0.859 at $\alpha = 1$ demonstrates that domain knowledge and data-driven inference provide complementary rather than redundant information. The fact that DPI alone ($\alpha = 0$) already achieves 67% directional agreement with LiNGAM validates the DPI as a meaningful asymmetric statistic, while the additional 20+ percentage points from domain knowledge highlight the value of expert input when available and reliable.

11.4 Examination of Assumptions

The theoretical results in this paper rely on several assumptions whose scope and relaxability merit discussion.

Additive noise model (ANM). The partial identifiability theorem (Theorem 29) assumes an ANM data-generating process. While this is weaker than LiNGAM’s joint requirement of linearity *and* non-Gaussianity, it still excludes important model classes: multiplicative noise, heteroscedastic models, and post-nonlinear models. In practice, the DPI’s ensemble design (averaging three asymmetric components) provides empirical robustness beyond the ANM setting, but formal guarantees for broader model classes remain an open problem.

Tree DAG structure. The phase–direction correspondence proof (Appendix A) assumes a tree DAG with positive edge weights. Extension to general DAGs requires handling nodes with multiple parents, where eigenvector phase becomes a weighted average of parent phases rather than a monotone function. Preliminary numerical experiments suggest the

correspondence holds approximately for sparse DAGs with bounded in-degree, but a formal proof is lacking.

Fixed charge parameter q . The spectral analysis depends on the charge parameter q , which controls the trade-off between symmetric coupling information ($q = 0$) and directional phase information ($q > 0$). Our experiments use $q = 0.25$ for eigenvector analysis and $q = 0.15$ for CCI computation (to avoid SCC degeneracy at $q = 0.25$). A principled q -selection procedure—analogue to bandwidth selection in kernel methods—would strengthen the practical applicability of the framework. Cross-validation against an external criterion (e.g., LiNGAM agreement rate or bootstrap stability) is a promising direction.

Domain knowledge quality. The framework assumes access to a domain knowledge matrix C_{domain} whose entries are non-negative and roughly calibrated to causal influence strength. While Theorem 23 shows that only the sign structure matters for r_{gradient} , the SCC and SCD values *do* depend on the magnitudes. Sensitivity analysis of the spectral indices to perturbations in C_{domain} would provide confidence bounds and is a natural extension.

11.5 Scalability

The magnetic Laplacian eigendecomposition is $O(n^3)$; for large n , randomized SVD achieves $O(nk^2)$ for the top k eigenpairs. The Hodge decomposition is $O(|E|)$ for sparse graphs. The DPI computation is $O(n^2 \cdot N)$ for N observations.

11.6 Limitations

1. The UCI Heart Disease experiment involves only 5 variables; validation on larger datasets (MIMIC-IV, Japanese health checkup cohorts with $n > 10^5$) is needed.
2. The charge parameter q significantly affects results; principled selection criteria (cross-validation against LiNGAM, stability analysis) require further development.
3. Full identifiability (Phase 3) remains an open problem.

11.7 Future Directions

1. Phase 2 identifiability proof for tree DAG + linear SEM.
2. ECD validation on MIMIC-IV and large-scale cohort data.
3. Temporal extension: time-lagged utility graphs with eigentrajectory extraction.
4. Automated p_{flip} estimation from LiNGAM agreement rates.
5. Data-adaptive α via bootstrap tests of asymmetric matrix consistency with correlation structure.
6. Formal proof that the phase-transition threshold $p_{\text{flip}}^* \approx 0.15$ generalizes beyond the 5-variable case.

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[To be completed upon submission.]

Appendix A. Proof of Phase–Direction Correspondence (Theorem 7)

We prove Theorem 7 in two stages: first for path graphs (Lemma 33), then for tree DAGs (Theorem 34) via structural induction.

Assumption 32 $G = (V, E)$ is a rooted tree DAG with n nodes. The structural equation model is $X_j = \sum_{i \in \text{pa}(j)} \beta_{ij} X_i + \varepsilon_j$ with $\beta_{ij} > 0$ for all $i \in \text{pa}(j)$. The utility graph assigns edge weights $w_{ij} = (U(i, j) + U(j, i))/2 > 0$ and direction signs $\sigma_{ij} = +1$ for each parent–child edge $i \rightarrow j$.

A.1 Stage 1: Path Graphs

Lemma 33 (Phase monotonicity on path graphs) Let $P = v_1 \rightarrow v_2 \rightarrow \cdots \rightarrow v_n$ be a directed path graph with uniform weight $w > 0$. Set $q = 0.25$. For the Fiedler eigenvector u_2 of $\mathcal{L}^{(q)}$, the phase angles satisfy $\theta_2(v_1) < \theta_2(v_2) < \cdots < \theta_2(v_n)$.

Proof At $q = 0.25$, the Hermitian adjacency matrix satisfies $H_{v_k, v_{k+1}}^{(q)} = iw$ and $H_{v_{k+1}, v_k}^{(q)} = -iw$. For the path graph, every interior node v_k ($2 \leq k \leq n-1$) has degree $d_{v_k} = 2w$; the endpoints have degree $d_{v_1} = d_{v_n} = w$. The normalized magnetic Laplacian row for interior node v_k gives

$$(1 - \lambda)u(v_k) = \frac{-iw}{d_{v_k}} u(v_{k-1}) + \frac{iw}{d_{v_k}} u(v_{k+1}), \quad (13)$$

where we suppress the eigenvector index and write $u = u_2$. Writing $u(v_k) = r_k e^{i\theta_k}$ and substituting into (13), the imaginary part of the equation (after dividing by r_k) gives

$$0 = \frac{w}{d_{v_k}} \left[\frac{r_{k+1}}{r_k} \sin(\theta_{k+1} - \theta_k + \frac{\pi}{2}) + \frac{r_{k-1}}{r_k} \sin(\theta_{k-1} - \theta_k - \frac{\pi}{2}) \right].$$

Using $\sin(\alpha + \pi/2) = \cos \alpha$ and $\sin(\alpha - \pi/2) = -\cos \alpha$:

$$\frac{r_{k+1}}{r_k} \cos(\theta_{k+1} - \theta_k) = \frac{r_{k-1}}{r_k} \cos(\theta_{k-1} - \theta_k).$$

For the Fiedler eigenvector with λ_2 small and positive, the amplitudes r_k vary smoothly and are bounded away from zero on a connected graph. The real part of (13) yields

$$(1 - \lambda_2)r_k = \frac{w}{d_{v_k}} \left[r_{k+1} \sin(\theta_{k+1} - \theta_k) + r_{k-1} \sin(\theta_k - \theta_{k-1}) \right].$$

Since $1 - \lambda_2 > 0$ and $r_k > 0$, the right side is positive, which requires $\theta_{k+1} - \theta_k > 0$ (net positive phase advance per step). For uniform weights, the symmetry of the recursion yields a constant phase increment

$$\Delta\theta = \theta_{k+1} - \theta_k = \arctan\left(\frac{2\pi q \cdot \tilde{w}}{1 + \tilde{w}^2}\right) > 0, \quad (14)$$

where $\tilde{w} = w/d_{v_k}$ is the normalized weight. Since each $\Delta\theta > 0$, we have $\theta_2(v_1) < \theta_2(v_2) < \cdots < \theta_2(v_n)$. $\blacksquare$

A.2 Stage 2: Extension to Tree DAGs

Theorem 34 (Phase–direction correspondence on tree DAGs) *Under Assumption 32 with $q = 0.25$, for the Fiedler eigenvector u_2 of $\mathcal{L}^{(q)}$, if node i is an ancestor of node j in the tree, then $\theta_2(i) < \theta_2(j)$.*

Proof By structural induction on the depth of the tree.

Base case. A tree of depth 1 is a star graph $r \rightarrow \{c_1, \dots, c_m\}$. The eigenequation at the root r (with $d_r = mw$) gives

$$(1 - \lambda)u(r) = \frac{1}{m} \sum_{j=1}^m iw \cdot \frac{u(c_j)}{\sqrt{w \cdot d_{c_j}}} \cdot \sqrt{\frac{d_{c_j}}{mw}}.$$

For leaf c_j with $d_{c_j} = w$:

$$(1 - \lambda)u(c_j) = \frac{-iw}{\sqrt{w \cdot mw}} u(r) = \frac{-i}{\sqrt{m}} u(r).$$

Writing $u(r) = r_0 e^{i\theta_0}$, we obtain $u(c_j) = \frac{r_0}{\sqrt{m(1-\lambda)}} \cdot e^{i(\theta_0 - \pi/2)}$, so $\theta_2(c_j) = \theta_0 - \pi/2 + \pi = \theta_0 + \pi/2 > \theta_0$ (modulo 2π , choosing the principal branch where all phases lie in $(-\pi, \pi]$). Hence $\theta_2(r) < \theta_2(c_j)$ for all children.

Inductive step. Assume the result holds for all subtrees of depth $\leq d$. Consider a tree of depth $d+1$ rooted at r . Let $c_1, \dots, c_m$ be the children of r , each heading a subtree T_j of depth $\leq d$. The eigenequation at r gives $u(r)$ in terms of $u(c_1), \dots, u(c_m)$ with phase offset $+\pi/2$ per edge (from the $i \cdot w$ factor in $H^{(q)}$), yielding $\theta_2(r) < \theta_2(c_j)$ by the same calculation as the base case. Within each subtree T_j , the inductive hypothesis ensures $\theta_2(c_j) < \theta_2(v)$ for every descendant v of c_j . Transitivity gives $\theta_2(r) < \theta_2(c_j) < \theta_2(v)$.

Perturbation argument for non-uniform weights. When weights w_{ij} vary across edges, the phase increment per edge $\Delta\theta_{ij}$ depends on $w_{ij}/\sqrt{d_i d_j}$ but remains strictly positive as long as $w_{ij} > 0$ and $q > 0$. The normalized weight ratio $\tilde{w}_{ij} = w_{ij}/\sqrt{d_i d_j} \in (0, 1]$ ensures $\Delta\theta_{ij} \in (0, \pi/2]$ by (14), preserving the strict monotonicity of phase along every root-to-leaf path. ■

Remark 35 *The restriction to tree DAGs is essential: in a general DAG with merging paths (“V-structures”), the phase contributions from different paths to a common descendant may interfere, potentially weakening the monotonicity. Empirically, the correspondence remains strong even for dense graphs (Section 7), but extending the formal proof beyond trees remains open.*

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